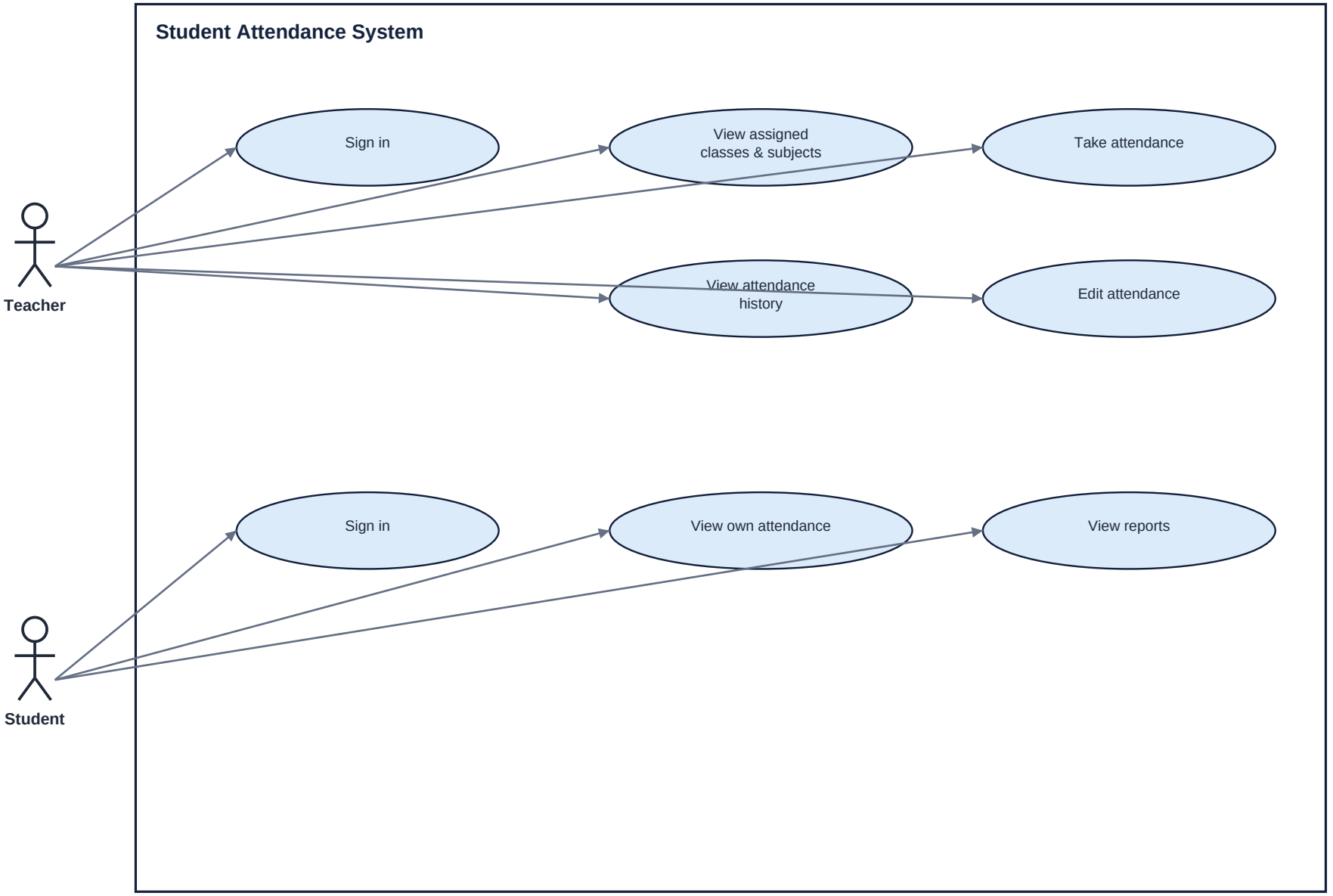
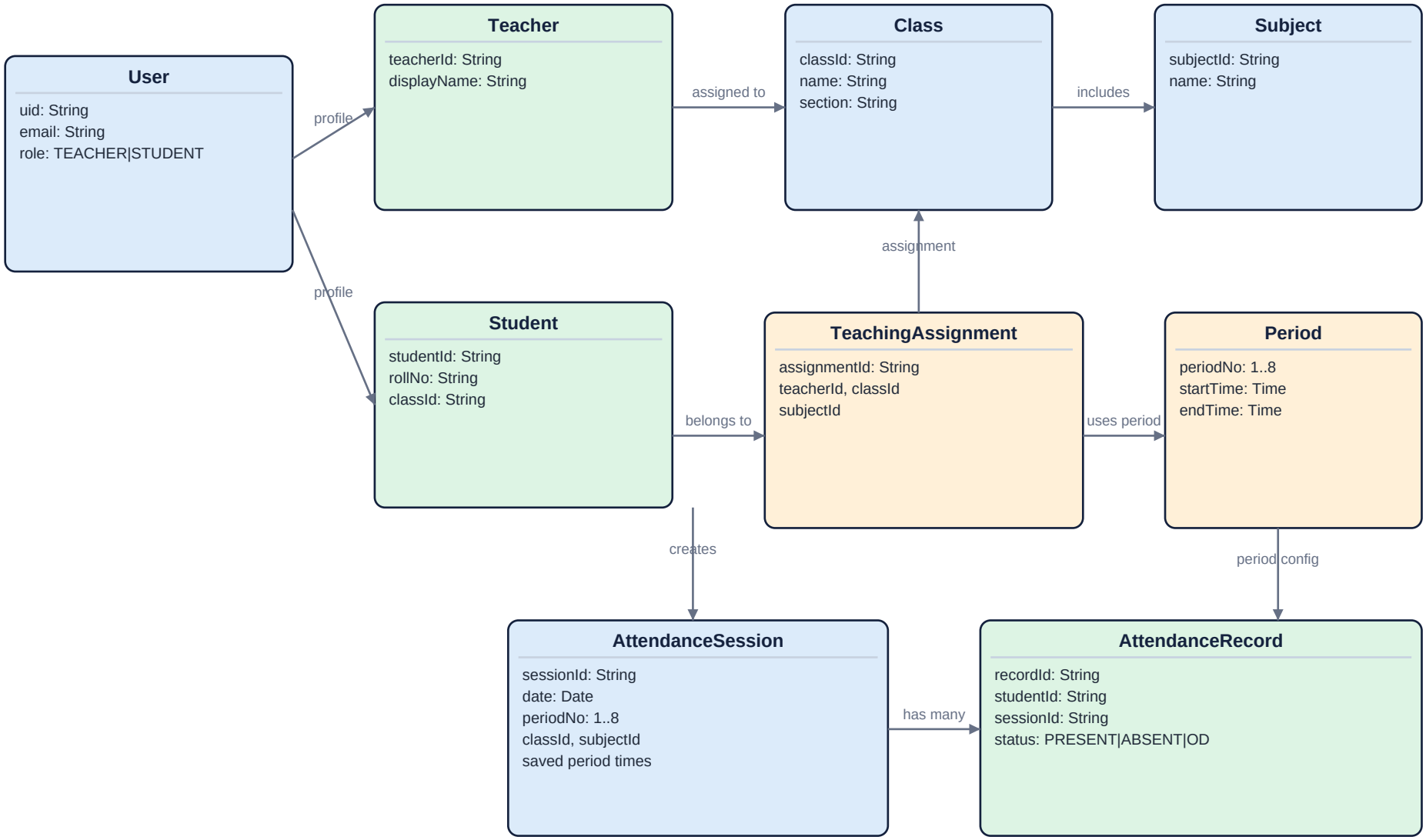
1. UML Use Case Diagram



Roles are limited to TEACHER and STUDENT. Teachers manage attendance only for authorized classes/subjects.

2. UML Class / Domain Diagram

Core entities and relationships

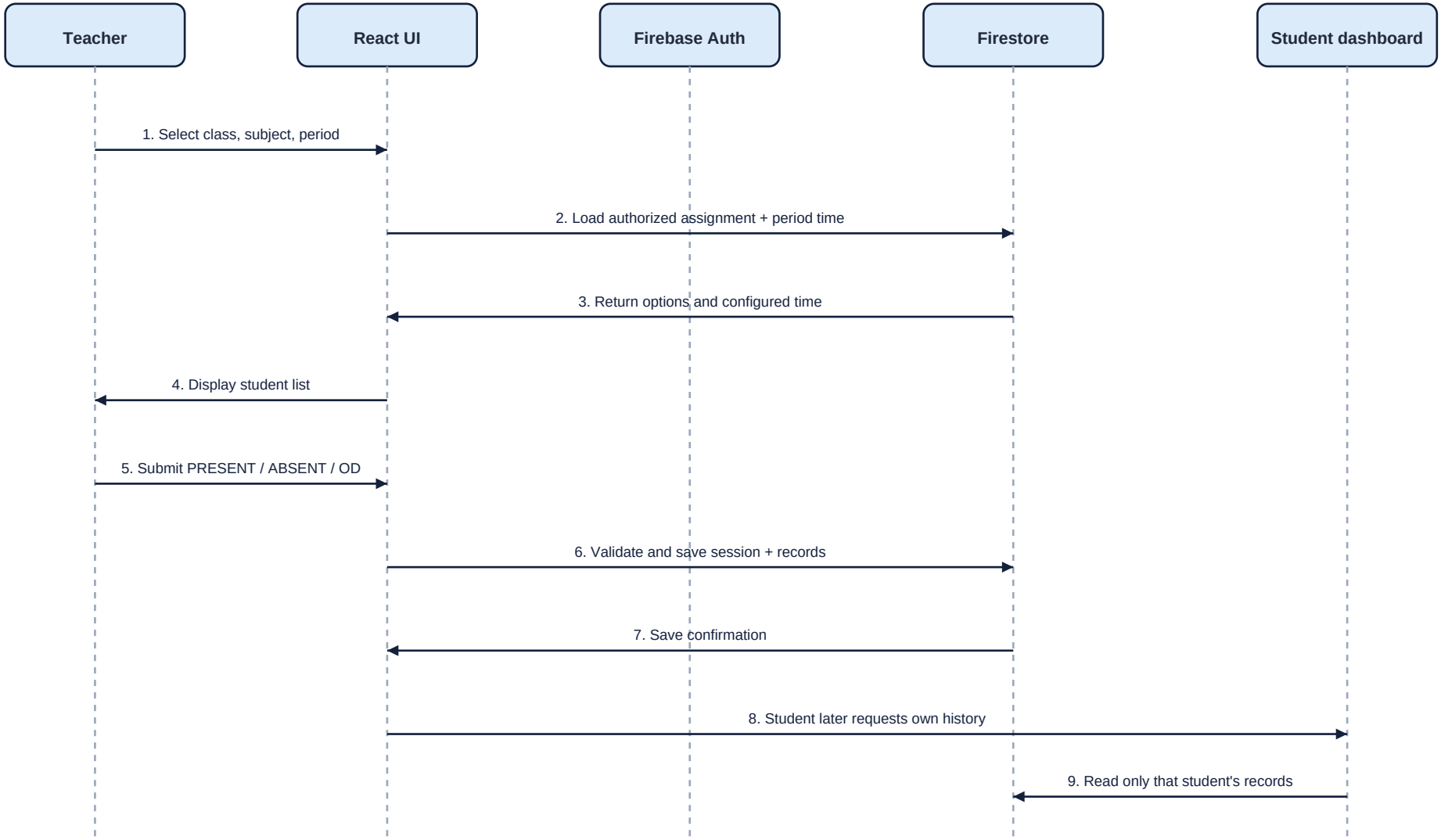


Conceptual domain model. Firebase Authentication stores credentials; Firestore stores profiles and application data.

Attendance percentage = (PRESENT + OD) / total conducted periods × 100.

3. UML Sequence Diagram

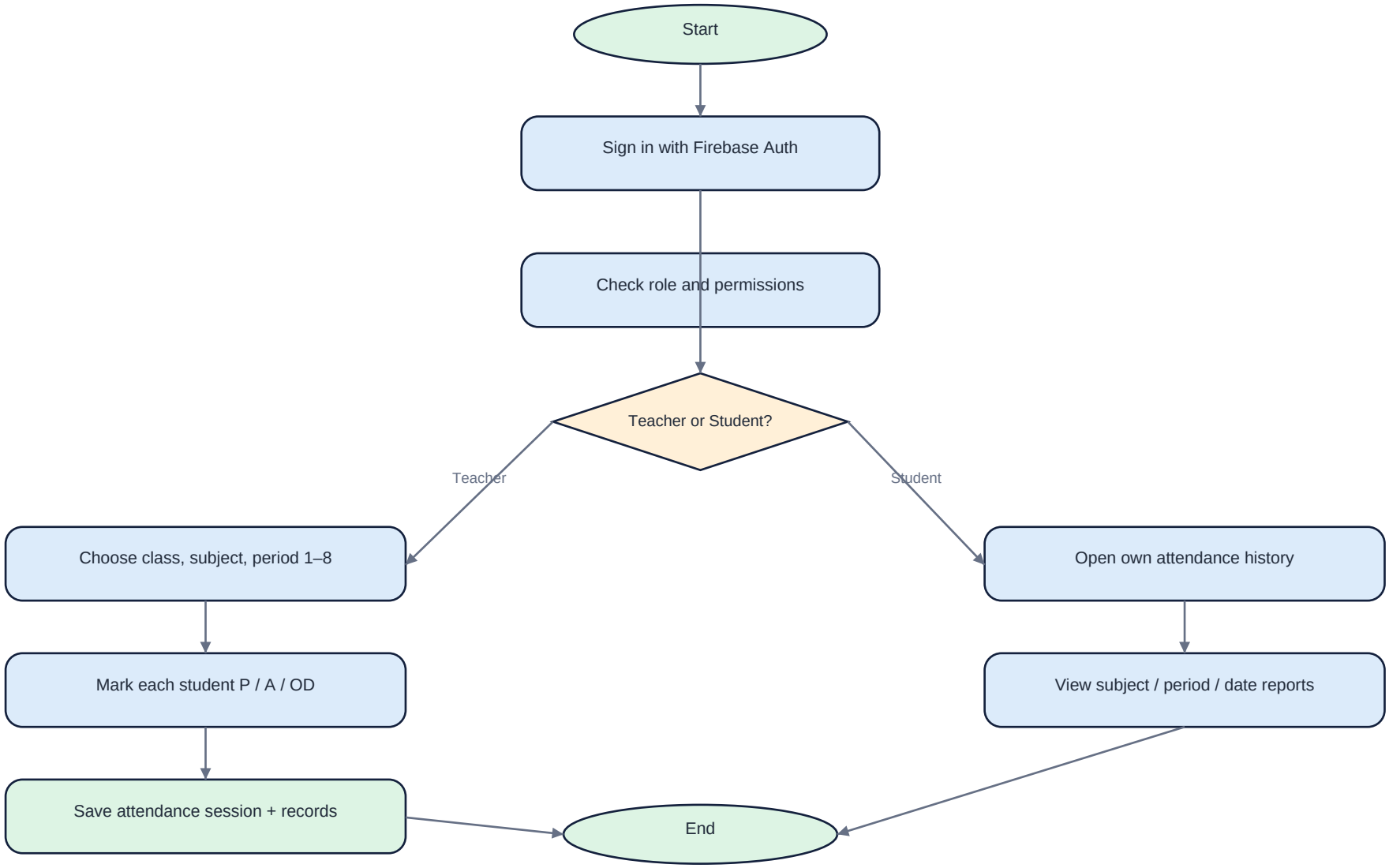
Teacher records attendance



Access control must enforce teacher assignment scope and student self-only reads.

4. UML Activity Diagram

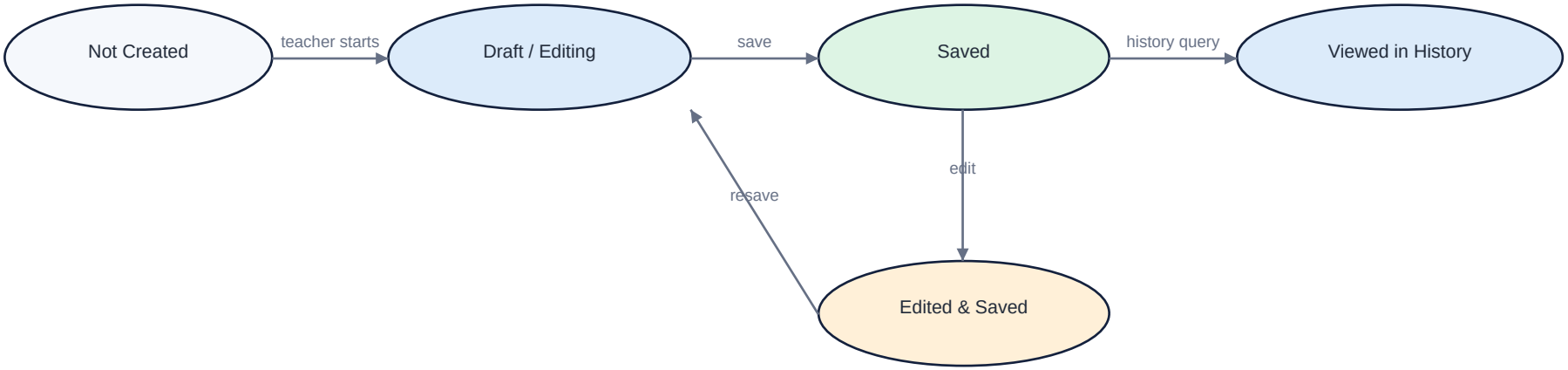
Attendance workflow



Period times are loaded automatically from configured period data; teachers do not enter custom start/end times.

5. UML State Diagram

Attendance record lifecycle



Valid status values: PRESENT, ABSENT, OD

OD is counted as attended for percentage calculations.

Only authorized teachers can create/edit records; students can view their own records only.